

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Hemalakshmi H - 192571049 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

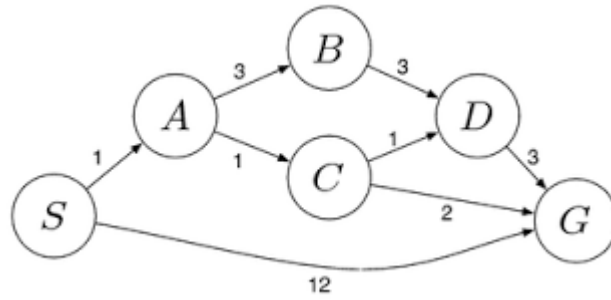
Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

1 Answer:

- To obtain exactly 2 gallons in the 4-gallon jug, we use a sequence of allowed operations: filling a jug completely, emptying a jug, and pouring water from one jug to another until one becomes full or empty.
 - First, fill the 3-gallon jug completely and pour all 3 gallons into the 4-gallon jug. Then fill the 3-gallon jug again. Pour from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug becomes full. Since the 4-gallon jug already contains 3 gallons, it can take only 1 more gallon, leaving exactly 2 gallons remaining in the 3-gallon jug. Now empty the 4-gallon jug completely. Finally, pour the 2 gallons from the 3-gallon jug into the empty 4-gallon jug.
 - At the end of these steps, the 4-gallon jug contains exactly 2 gallons of water.
-

2. Answer:

i. Percepts for the Mars Rover

The percepts include all sensor inputs the rover receives from its environment: visual images from cameras, chemical readings from spectrometers, soil composition data, temperature, atmospheric conditions, terrain information, obstacle detection, wheel-slip data, and positional information from onboard navigation sensors.

ii. Operating Environment

The operating environment is partially observable, dynamic, continuous, unpredictable, and hostile. Mars has rough terrain, dust storms, extreme temperatures, communication delays with Earth, and no real-time human intervention. The rover must operate autonomously in an outdoor, open, and hazardous environment with limited energy resources.

iii. Actions the Agent Can Take

The rover can move forward, backward, or turn; navigate around obstacles; capture images; drill or scoop soil; analyze samples using onboard instruments; store or transmit collected data; adjust solar panels; and perform self-diagnostics or system checks.

iv. Evaluating the Performance of the Agent

Performance can be evaluated based on accuracy of scientific measurements, successful navigation without damage, efficiency in energy usage, number of samples collected and analyzed, reliability of communication with Earth, ability to avoid hazards, and overall mission objectives achieved within time and resource limits.

v. Suitable Agent Architecture and Why

A hybrid agent architecture (combining reactive and deliberative components) is most suitable. The reactive layer allows quick responses to hazards like obstacles or wheel slip, while the deliberative layer enables long-term planning for navigation, sample analysis, and mission goals. This architecture handles both real-time decisions and complex scientific tasks effectively in an uncertain environment like Mars.

3. Answer:

- **State space:** A state is a configuration of queens on the board. A common representation is: each state corresponds to placing queens column by column, with one queen per column. Then a state can be represented as an 8-tuple, where the i -th element gives the row position of the queen in column i . The full state space consists of all possible ways of placing queens (intermediate states may have fewer than 8 queens).
- **Initial state:** Typically, the initial state is an empty board (no queens placed), or a board with the first queen placed in some fixed position (e.g., first column, any row). From there, queens are added one by one.
- **Successor function (actions):** The action is “place a queen in the next column in any row that is not attacked by already placed queens.” From a partial configuration with k queens placed in the first k columns, the successor states are all valid ways of placing a queen in column $k + 1$ such that it is not in the same row or diagonal as any existing queen.
- **Goal test:** A state is a goal state if 8 queens have been placed on the board and no two queens attack each other. Formally, the configuration must satisfy:
 - All 8 columns contain exactly one queen.
 - No two queens share the same row.
 - No two queens share the same diagonal.
- **Path cost:** Often, all actions have the same cost (e.g., cost = 1 per queen placement), so the path cost is simply the number of queens placed. Since any solution with 8 queens is acceptable, path cost is usually not critical; we mainly care about finding any valid solution.
- **Search strategies:** Common search strategies for exploring the search space include:
 - Depth-first search (DFS): place queens column by column, backtracking when a conflict occurs.
 - Backtracking search with constraint checking: at each step, prune branches where the new queen conflicts with existing queens.
 - Heuristic search (e.g., hill climbing, local search): start from a complete but possibly attacking configuration and move queens to reduce the number of conflicts. These strategies systematically explore or improve configurations until a non-attacking arrangement is found.

Thus, the 8-Queens problem is formulated as a search problem with clearly defined states, initial state, successor function, and goal test, and solved by exploring the search space using appropriate search strategies.

4. Answer:

- A customer using the OLA Cab booking application needs to choose a cab type (mini, micro, sedan, shared, prime, etc.) and book a ride from a source location to a destination. This requires an agent that can make decisions based on user preferences, available cab options, and real-time constraints.
- **Type of Problem-Solving Agent**
A **goal-based agent** is suitable for this problem. The agent's goal is to successfully book a cab that satisfies the customer's requirements (cab type, availability, cost, and travel time). It evaluates different options and selects the best one to achieve the goal.

Pseudocode for Cab Booking Agent:

CabBookingAgent()

Input: source_location, destination_location, preferred_cab_type

Output: confirmed_booking or failure_message

Begin

available_cabs $\leftarrow$ GetAvailableCabs(source_location)

if available_cabs is empty then

return "No cabs available"

end if

filtered_cabs $\leftarrow$ FilterByCabType(available_cabs, preferred_cab_type)

if filtered_cabs is empty then

filtered_cabs $\leftarrow$ available_cabs // fallback to any cab

end if

best_cab $\leftarrow$ SelectCabWithMinimumETA(filtered_cabs)

fare $\leftarrow$ CalculateFare(best_cab, source_location, destination_location)

booking_status $\leftarrow$ BookCab(best_cab, source_location, destination_location, fare)

if booking_status = success then

return "Cab booked successfully: " + best_cab

else

return "Booking failed. Try again."

end if

End

5 Answer:

To determine the least-cost path from the starting warehouse **S** to the destination warehouse **G**, the Uniform Cost Search (UCS) algorithm explores the delivery network by always expanding the node with the lowest cumulative travel cost.

The weighted graph contains the following routes and costs:

- S $\rightarrow$ A : 1
- A $\rightarrow$ B : 3
- A $\rightarrow$ C : 1
- B $\rightarrow$ D : 3
- C $\rightarrow$ D : 1
- C $\rightarrow$ G : 2
- D $\rightarrow$ G : 3
- S $\rightarrow$ G : 12

UCS begins at **S** with cost 0 and expands nodes in increasing order of path cost.

Step-by-step UCS exploration:

1. Start at **S** (cost = 0). Frontier = {A (1), G (12)}
2. Expand **A** (lowest cost = 1). New frontier = {C (1+1=2), B (1+3=4), G (12)}
3. Expand **C** (lowest cost = 2). New frontier = {G (2+2=4), B (4), D (2+1=3), G (12)}
4. Expand **D** (lowest cost = 3). New frontier = {G (4), G (3+3=6), B (4), G (12)}
5. Expand **G** with cost **4** (lowest among all G entries). Since G is the goal, UCS stops here.

Least-cost path found by UCS: $S \rightarrow A \rightarrow C \rightarrow G$

Total cost: $1 (S \rightarrow A) + 1 (A \rightarrow C) + 2 (C \rightarrow G) = 4$